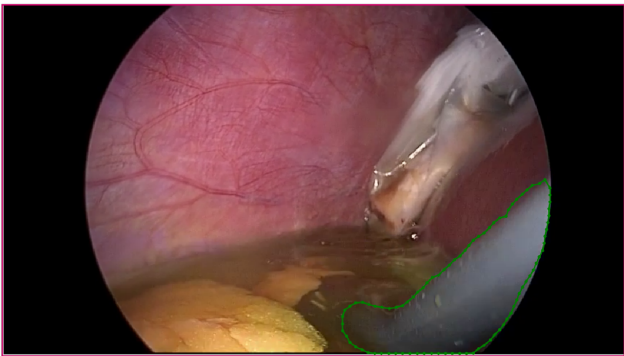
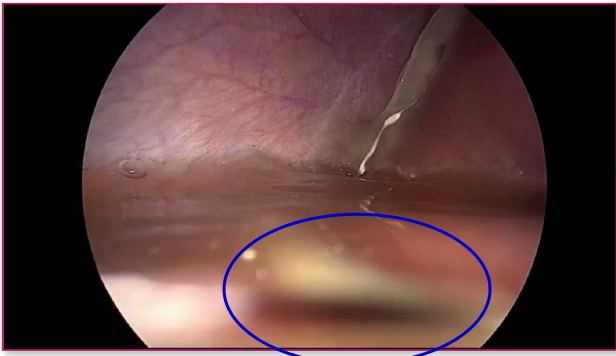


Annotation guidelines for hard cases in CholecInstanceSeg

2 List of Figures

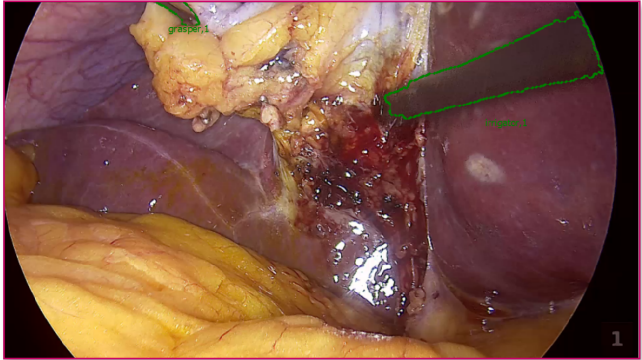
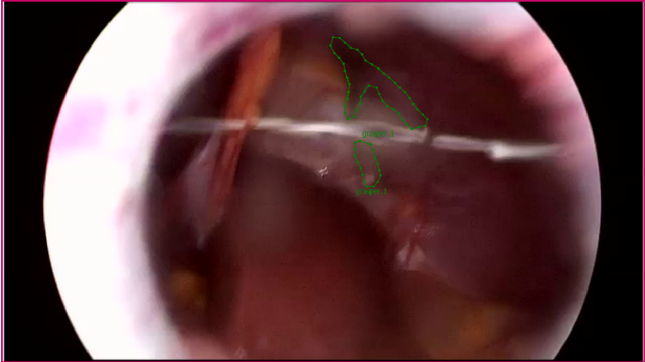
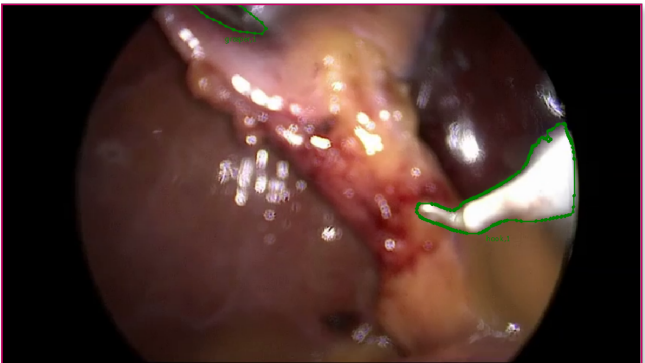
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Tool in fluid scenarios

Scenario	Example
Refraction by fluid: <i>annotate refracted pixels</i>	 The green outline shows the tool's annotated refracted pixels.
Occlusion by fluid: <i>do not annotate</i>	 The blue circle highlights the tool that is fully occluded by fluid and should not be annotated.

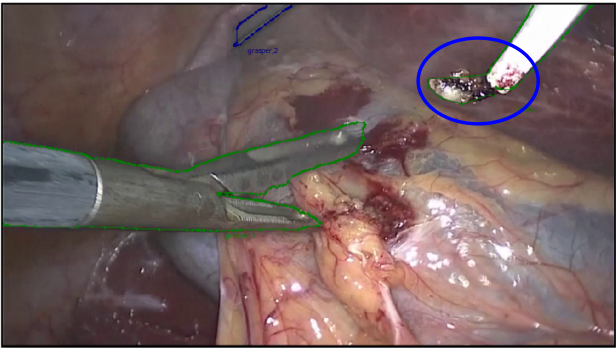
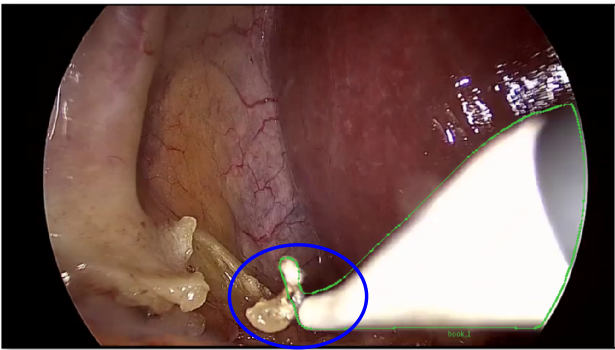
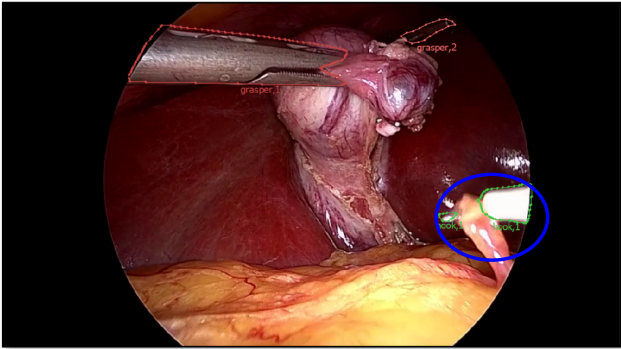
Supplementary figure 1. Annotation guidelines for tools in fluid scenarios

Blur scenarios

Scenario	Example
High visibility blur: <i>annotate based on the expected shape of the instrument</i>	 The green outline shows the annotation for the blurred tool which is mostly visible
Low visibility blur: <i>use neighbouring frames to infer the tool's shape and annotate accordingly.</i>	 The green outline shows the annotation for the blurred tool with low visibility
Out-of-focus blur: <i>annotate with the inferred shape of the tool in mind.</i>	 The green outline shows the annotation for a tool out focus

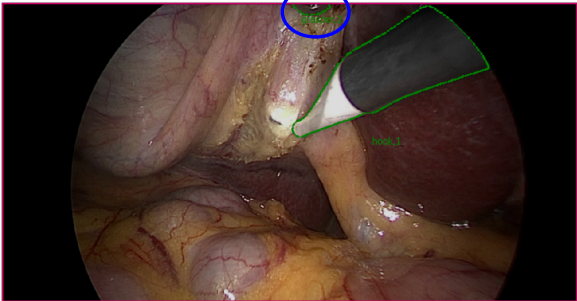
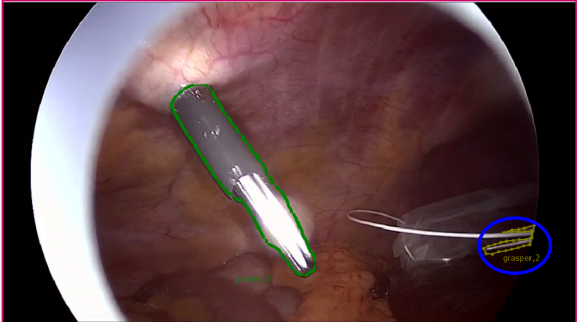
Supplementary figure 2. Annotation guidelines for blur scenarios.

Tissue attached to tool tip scenarios

Scenario	Example
Small amount of tissue attached to tool tip: <i>annotate the tissue as part of the tool.</i>	 The blue circle highlights a tool tip annotated with a small amount of tissue.
Tissue attached to the tool tip, but the tip is clearly visible: <i>annotate only the tool tip; do not annotate the tissue.</i>	 The blue circle highlights a tool tip annotated with the tip only, as it is clearly visible.
Tissue fully occludes parts of the tool tip: annotate as as occlusion; do not annotate occluded portions	 The blue circle highlights a tool tip that is fully occluded by tissue. It is annotated as occlusion

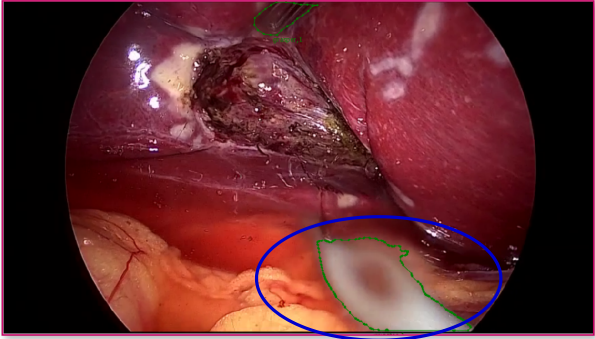
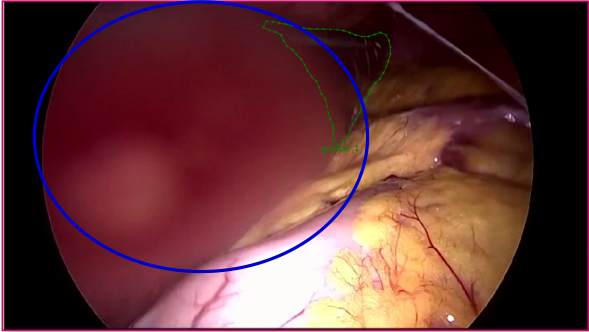
Supplementary figure 3. Annotation guidelines for tissue attached to tool tip scenarios.

Tools at the edge of the image or far away from the camera

Scenario	Example
Tool at the edge of the frame: <i>annotators must ensure these tools are always annotated despite their partial visibility</i>	 The blue circle highlights an instrument partially visible at the edge of the image. Annotators should be vigilant for instruments entering or exiting the frame.
Tools distant from the camera: <i>tools that are situated far from the camera and might be easily overlooked; annotators must ensure these instruments are always annotated</i>	 The blue circle highlights an instrument distant from the camera. Annotators should take note of instruments in the background to ensure comprehensive annotation.

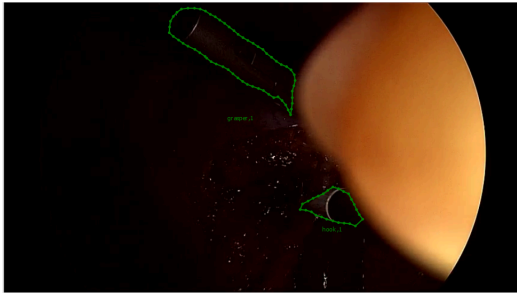
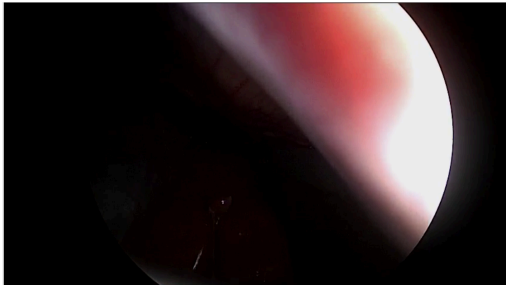
Supplementary figure 4. Annotation guidelines for instruments at the edge of the frame or distant from the camera.

Lens dirtiness scenarios

Scenario	Example
Lens dirtiness refraction: <i>annotate all the refracted pixels.</i>	 The blue circle highlights a tool with pixels affected by lens dirtiness, causing light refraction. Annotators should include all visible pixels distorted by the refraction in the annotation.
Lens dirtiness occlusion: <i>do not annotate occluded pixels</i>	 The blue circle highlights a tool partially obscured by lens dirtiness, resulting in occlusion. Annotators should exclude these occluded areas from annotation.

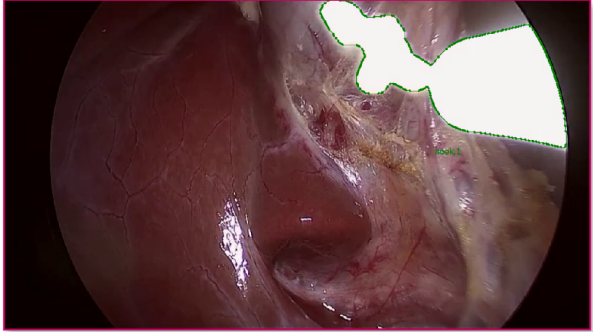
Supplementary figure 5. Annotation guidelines for tools affected by lens dirtiness.

Low lighting and saturated lighting scenarios

Scenario	Example
Low lighting, tool partially visible: <i>annotate based on the expected shape of the instrument.</i>	 The green outline shows annotations for a partially visible tool. Annotate the visible pixels, considering the instrument's expected shape.
Low lighting, tool not visible: <i>do not annotate</i>	 The image has low visibility such that instruments visible in previous frames are completely obscured. Do not annotate unseen instruments.
Saturated lighting, tool not visible: <i>do not annotate</i>	 The image has low visibility due to saturated lighting, making the instruments indiscernible. Do not annotate.

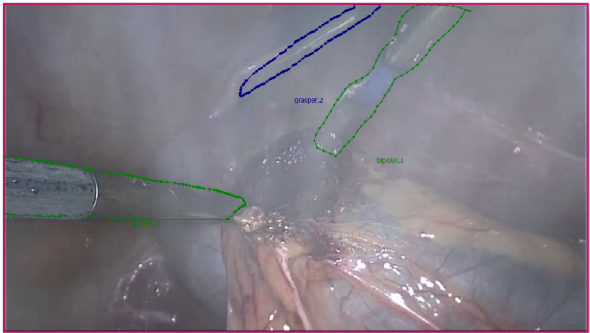
Supplementary figure 6. Annotation guidelines for tools in low lighting and saturated lighting.

Light reflection of tools

Scenario	Example
Reflection of light of a tool: <i>annotate based on the expected shape of the instrument.</i>	 The green outline shows annotations of a tool which is reflecting light.

Supplementary figure 7. Guidelines for annotating tools under light reflection conditions.

Low visibility due to smoke

Scenario	Example
Low visibility due to smoke: <i>annotate based on the expected shape of the instrument.</i>	 Annotators are advised, to annotate based on the expected shape of the instrument when smoke leads to low visibility

Supplementary figure 8. Guidelines for annotating tools during low visibility due to smoke.

More Evaluation metrics for baseline models